

Can barium stars form by stable mass transfer in AGB binaries?

At least half of low- and intermediate-mass stars exist in binary systems, and a large fraction of these interact with their companion star during the asymptotic giant branch (AGB) phase of evolution. The products of such interaction are common and varied, but of particular interest are *barium- and carbon-enriched stars* (see e.g. the review by Mathieu & Pols 2025). These are relatively unevolved (MS or RGB) stars with enhanced abundances of C, Ba, and other s-process elements, which they acquired through mass transfer from a former AGB companion in which they were synthesized, and which is currently a white dwarf. Here we collectively call these stars ‘Ba stars’, although they also comprise differently-named groups (such as CH stars and CEMP stars),

Ba stars are found in binaries with orbits that are often eccentric and have periods between about 100 and 20,000 days. These orbital properties are difficult to explain: AGB stars reach radii of several AU, causing them to fill their Roche lobes in orbits less than ~ 3000 days, and strong tides are expected to circularise these orbits. Moreover, RLOF from giants including AGB stars has long been considered to be unstable, leading to a common-envelope phase that allows little accretion and drastically shortens the orbit to much less than 100 days. Thus, the ubiquity of eccentric Ba stars with periods of hundreds to thousands of days is surprising.

Recent detailed binary evolution modelling has shown that RLOF from RGB and early-AGB donor stars can be stable over a much larger range of periods and mass ratios than what has been assumed based on simplified models (e.g. see Ge+2020, Temmink+2023). This suggests the possibility that *stable mass transfer from thermally pulsing (TP) AGB donors*, which are the source of the abundances seen in Ba stars, may be responsible for the Ba stars with orbits up to a few thousand days. However, this remains to be verified with detailed binary modelling, which so far does not extend into the regime of TP-AGB donors, apart from a few case studies (Pastetter+1989, Sun+2023). Mass transfer from TP-AGB stars is complicated by the presence of thermal pulses which cause rapid variations in stellar radius.

The goal of this project is to explore whether (and under which circumstances) RLOF from TP-AGB donor stars can indeed be stable, and whether the orbits of some of the observed Ba stars can be reproduced in this way. The project will be conducted with the stellar evolution code MESA, which has been successfully used both for evolving detailed models of single TP-AGB stars (Cinquegrana+2022, Rees+2024a,b), and for studies of mass-transfer stability from pre-TP-AGB giant donors (Temmink+2023). The challenge of this project is to combine both approaches to model mass transfer in TP-AGB binaries.

Project planning:

1. Literature study, consisting of the references listed below.
2. Detailed single TP-AGB modelling:
 - a) Gain familiarity with using MESA for AGB modelling, following the preliminary work by Casper Moltzer. and the papers by Cinquegrana+2022 and Rees+2024.
 - b) Construct detailed single TP-AGB evolution models, for a chosen mass, to serve as starting points for binary modelling.
3. Binary mass transfer modelling:
 - a) Follow the approach of Temmink+2023, both for setting up the binary models in MESA, and for checking for mass transfer stability.
 - b) For a consistency check, first try to reproduce some results by Temmink+2023 for E-AGB donors.
 - c) Since we want to relax the assumption of conservative mass transfer, explore the MESA options to allow for mass and angular-momentum loss from the system. First apply to an E-AGB donor, monitoring the effect on the orbital evolution.
 - d) Subsequently, choose a more evolved starting model and a wider orbit to model TP-AGB mass transfer. Model a situation most likely to be stable, i.e. a mass ratio close to unity.
 - e) When successful, extend calculations to different initial periods and mass ratios.
4. Compare the results after stable mass transfer to the observed properties of Ba stars. (In practice, this step may be simultaneous with the last steps of part 3 for choosing optimal modelling parameters.)
5. Draw conclusions and write up the results in a Master thesis.

References:

- Cinquegrana, Joyce & Karakas, 2022, ApJ 939, 50
- Ge, Webbink, Chen & Han, 2020, ApJ, 899, 132
- Mathieu & Pols, 2025, ARAA 63, 467
- Pastetter & Ritter, 1989, A&A 214, 186
- Rees, Izzard & Karakas, 2024, MNRAS 527, 9643
- Rees & Izzard, 2024, MNRAS 531, 4033
- Temmink et al, 2023, A&A 669, A45
- Sun & Mathieu, 2023, ApJ 944, 89